

Television Exposure and Perceived Community Norms Regarding Domestic Violence in India

Shivam Gangwani

M.S. Quantitative Economics

Semester-II

Econometrics-I: Term Paper

Indian Statistical Institute, Delhi

June 2026

Abstract

This paper examines the relationship between television exposure and perceptions of community norms regarding intimate partner violence in India using data from the India Human Development Survey (IHDS). The outcome variable is an index constructed from women's reports of whether wife-beating is considered usual in their community under a range of circumstances. Using regression analysis with demographic and socioeconomic controls, the study investigates whether greater television exposure is associated with differences in these perceptions. The results indicate a negative association between television exposure and the perceived acceptance of wife-beating, suggesting that access to mass media may be linked to weaker perceptions of intimate partner violence as a common feature of community life.

Contents

1	Introduction	1
2	Data Source	1
3	Empirical Framework	2
3.1	Study Variables and Population	2
3.2	Empirical Model	2
4	Variable Definitions	2
4.1	Dependent Variable	2
4.2	Main Explanatory Variable	3
4.3	Controls	3
5	Descriptive Statistics	4
6	OLS Estimates	8
7	Heterogeneity and Robustness	11
7.1	Heterogeneity Analysis	11
7.2	Robustness Checks	12
8	Causality	13
9	Bound Analysis	13

List of Figures

1	Distribution of Norms Index	5
2	Mean Norms Index by State	6
3	Mean Norms Index by TV Frequency and Residence	7

List of Tables

1	Components of the Wife-Beating Norms Index	3
2	Summary Statistics	5
3	Mean Norms Index by Television Viewing Frequency	7
4	OLS Estimates: Television Viewing and Norms Index	10
5	Heterogeneity Analysis	11
6	Robustness Checks	12

1 Introduction

This paper examines whether television exposure is associated with women’s perceptions of community norms regarding intimate partner violence in India. Perceived norms shape expectations about what is normal, permissible, and likely to be tolerated. A woman who believes wife-beating is common in her community may be less likely to challenge it, report it, or seek help.

Perceptions of community norms are shaped by the social environment individuals are embedded in, and mass media is a plausible channel through which those perceptions evolve. Television exposes women to social environments beyond their immediate community. As a result, women may update beliefs about how common wife-beating is, even if their own personal attitudes remain unchanged. The direction of television’s influence, however, is theoretically ambiguous. While Hindi cinema increasingly portrays women in professional, autonomous, and non-traditional roles¹, Indian television has been dominated by soap operas,² often centered on traditional family structures and conventional gender roles. Television exposure may therefore either weaken or reinforce existing perceptions of gender norms - more specifically, whether wife-beating is a socially accepted feature of community life. Whether regular television viewing is associated with stronger or weaker perceptions of the community acceptance of intimate partner violence is ultimately an empirical question, and one this paper seeks to investigate.

2 Data Source

We use data from the India Human Development Survey-II (IHDS-II), conducted in 2011–12 by the National Council of Applied Economic Research (NCAER) and the University of Maryland. The IHDS-II is a nationally representative, multi-topic survey covering over 42,000 households across India, which provides detailed information on demographics, education, and household characteristics.

The relevant dataset is the Eligible Women data merged with the household data. As prescribed by the IHDS for joining these datasets, we perform a many-to-one merge using a set of unique identifiers: STATEID, DISTID, PSUID, HHID, HHSPLITID.

¹For instance, *Chak De! India* (2007) presents women as ambitious athletes competing at the national level, while *Salaam Namaste* (2005) portrays an independent, career-focused woman whose lifestyle challenges conventional norms regarding marriage and gender roles.

²For example, *Kyunki Saas Bhi Kabhi Bahu Thi* (2000–2008) and *Kahaani Ghar Ghar Kii* (2000–2008).

3 Empirical Framework

3.1 Study Variables and Population

The outcome of interest is the Wife-Beating Norms Index (referred to as the Norms Index from now), constructed as the mean of six binary indicators capturing whether a woman reports that it is usual in her community for a husband to beat his wife under a range of circumstances (detailed in Section 4.1). The primary explanatory variable is self-reported television viewing frequency—never, sometimes, or regularly—drawn from the India Human Development Survey (2011–12). The unit of observation is ever-married women aged 15 and above³, and the estimation sample comprises approximately 39,171⁴ women after excluding observations with missing values on key variables.

3.2 Empirical Model

The baseline estimating equation is

$$Y_i = \alpha + \beta_1 \text{Sometimes}_i + \beta_2 \text{Regularly}_i + \mathbf{X}_i' \gamma + \varepsilon_i$$

where Y_i is the Norms Index for woman i , and the television viewing dummies take *Never* as the (omitted) reference category.

$\mathbf{X}_i$ is a vector of controls described in Section 4.3. Standard errors are clustered at the PSU (village/neighbourhood) level, since both television access and prevailing gender norms are likely correlated within the same village or urban neighbourhood.

The coefficients of interest are β_1 and β_2 , which capture the conditional difference in the perceived community acceptance of wife-beating between women who watch television sometimes or regularly (respectively), relative to those who never watch television.

4 Variable Definitions

4.1 Dependent Variable

The Wife-Beating Norms Index Y_i is constructed as the mean of six binary indicators (GR34–GR39)⁵, each recording whether the respondent reports that it is usual in her community for a husband to

³Although the IHDS documentation describes eligible women as aged 15–49, the dataset contains a substantial number (more than 10% of the sample) of respondents above age 49. These observations are retained in the analysis.

⁴Less than 60 observations (under 0.2% of the sample) are dropped across specifications due to missing values. The proportion of missing observations is sufficiently small that any resulting sample-selection concerns are unlikely to be quantitatively important.

⁵The corresponding variable codes from the IHDS questionnaire are capitalised and mentioned in brackets where applicable

beat his wife under a specific circumstance:

Table 1: Components of the Wife-Beating Norms Index

Component	Mean
Going out without informing husband	0.513
Suspected infidelity	0.843
Natal family fails to provide expected gifts, money, or jewelry	0.325
Neglecting house or children	0.424
Not respecting family elders	0.475
Not cooking properly	0.319
Norms Index	0.483

Notes: Each component is a binary indicator equal to one if the respondent reports that wife beating is considered usual in the community under the stated circumstance, and zero otherwise. The Norms Index is constructed as the simple average of the six indicators.

4.2 Main Explanatory Variable

Television viewing frequency (MM3W) is a self-reported categorical variable with three levels: never (omitted), sometimes, and regularly. The variable is entered as two dummy variables, with women who never watch television serving as the reference category.

4.3 Controls

The controls used are described below, along with a brief justification for their inclusion in the specification:

- Age (EW6): Attitudes towards gender roles vary systematically across generations and age is also likely to influence patterns of television consumption.
- Own years of schooling (EW8): Education is a strong correlate of gender attitudes and also predicts household media access.
- Marital Status (RO6): It is included as a categorical control to account for differences in household circumstances and life-cycle experiences that may shape both media exposure and attitudes towards wife-beating. The omitted category is currently married women.
- $\ln(\text{consumption per capita})$ (COPC): Log consumption per capita is included as a proxy for household economic status, since wealthier households are both more likely to own televisions and may hold different attitudes independent of viewing behaviour.
- Urban/Rural residence (URBAN2011): Included as a binary control, since urban residence predicts both television access and exposure to a wider range of social attitudes.

-
- Caste and Religious group fixed effects (GROUPS): This is a seven-category fixed effect (GROUPS), with categories: Brahmin, Forward Caste, OBC, Dalit, Adivasi, Muslim, and Christian/Sikh/Jain. This is motivated by the well-documented variation in gender norms and social practices across these groups in India.
 - State fixed effects (STATEID): State fixed effects are included to absorb state-level variation in culture, institutions, and media environments⁶.

Note: Spouse's years of schooling (SPED6) is omitted from the above set of controls, even though we expect it to correlate with our dependent as well as primary explanatory variable. This is because the variable is observed only for women with a living spouse - controlling for this would restrict our sample to currently married women. Instead, we include this in a robustness specification later.

5 Descriptive Statistics

Table 1 presents summary statistics for the continuous variables. The Norms Index has a mean of 0.484 and a standard deviation of 0.337, indicating substantial variation across respondents. The distribution is U-shaped, with mass concentrated at the extremes. About 11% of women report that none of the six scenarios are usual in their community (index = 0), while 18% report that all six are usual (index = 1). The single largest share—around 22%—is at 1/6, indicating that only one scenario is perceived as usual. Intermediate values are less common. This pattern suggests that women differ substantially in their perceptions of community norms regarding wife-beating.

The sample skews rural, with 66% of women residing in rural areas. The median woman is 36 years old. Log consumption per capita has a mean of 9.90 and a standard deviation of 0.67, reflecting moderate economic heterogeneity across households.

Figure 2 highlights substantial geographic variation in the Norms Index across Indian states, providing preliminary motivation for the inclusion of state fixed effects in later specifications.

Education levels are low on average. Approximately 38% of women in the sample report no formal schooling, and the modal completed level is secondary education (10th class), reached by around 9% of respondents. Spousal education is somewhat higher — only 20% of husbands report no schooling — with secondary education again the most common completed level (13%). This pattern of wives being less educated than their husbands on average is consistent with broader evidence on educational gender gaps in India during this period.

⁶India has significant variation in gender norms (and therefore perceived gender norms) across states. For instance, while most of North India exhibits patriarchal norms, Northeast India has several prominent matrilineal societies such as the Khasi and Garo tribes of Meghalaya. These inter-state differences necessitates the inclusion of state-level fixed effects.

Table 2: Summary Statistics

Variable	Mean	Std. Dev.	Min	Max	N
Norms Index	0.484	0.337	0.00	1.00	39,357
Log consumption per capita	9.898	0.674	6.56	14.20	39,348
Age	36.33	9.868	15.00	81.00	39,357
Years of education (self)	5.18	4.92	0.00	16.00	39,357
Years of education (spouse)	6.58	5.09	0.00	16.00	39,357
Urban (=1)	0.340	0.473	0.00	1.00	39,357

Note: Sample comprises ever-married women aged 15 and above from the IHDS (2011). N varies slightly due to missing values on consumption expenditure. The proportion of missing values is 0.02% – too small relative to the sample size to be a cause for concern.

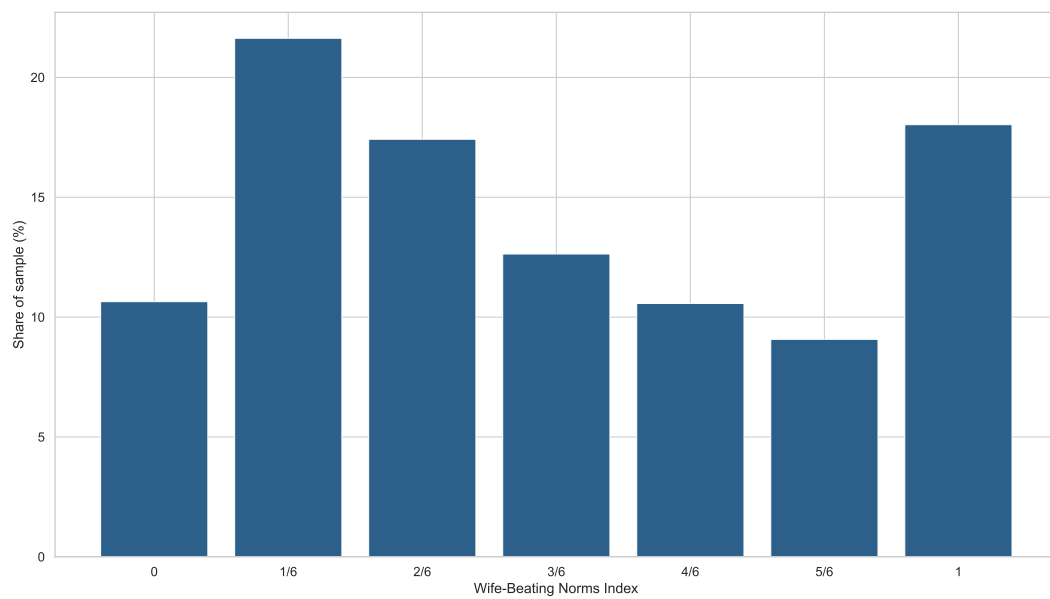
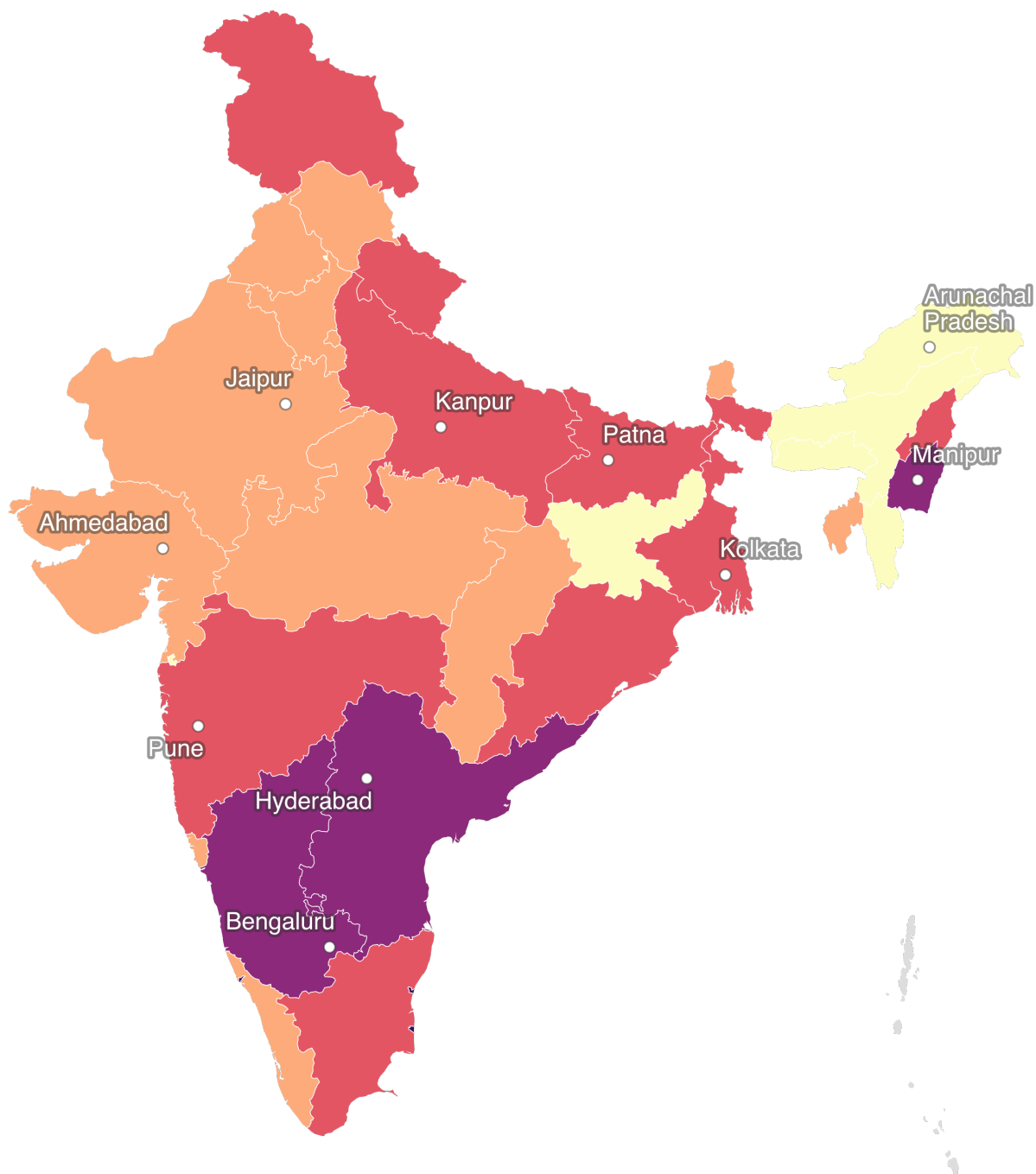
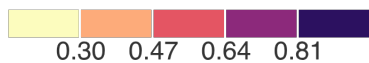


Figure 1: Distribution of Norms Index

Wife-Beating Norms Index



Created with Datawrapper

Figure 2: Mean Norms Index by State

On caste and religion, OBCs form the largest group at 34%, followed by Dalits (21%), Forward Castes (17%), Muslims (12%), and Adivasis (8%). Brahmins and Christians/Sikhs/Jains together account for roughly 8% of the sample.

Table 3 shows mean Wife-Beating Norms Index by television viewing frequency. Among women who never watch television, the mean index is 0.513; among sometimes-viewers it is 0.531; and among regular viewers it is 0.456. The difference in means is non-monotonic—sometimes-viewers have a higher mean than never-viewers—which likely reflects compositional differences across groups. Women who watch television more frequently are also more likely to be urban and educated, characteristics that may themselves be associated with lower perceptions that wife-beating is a usual feature of community life. The descriptive statistics therefore provide an initial picture of the relationship, but do not account for differences in respondent characteristics across viewing groups.

Table 3: Mean Norms Index by Television Viewing Frequency

Television Viewing	Mean	Std. Dev.	N
Never	0.513	0.335	7,325
Sometimes	0.531	0.338	8,853
Regularly	0.456	0.334	22,934

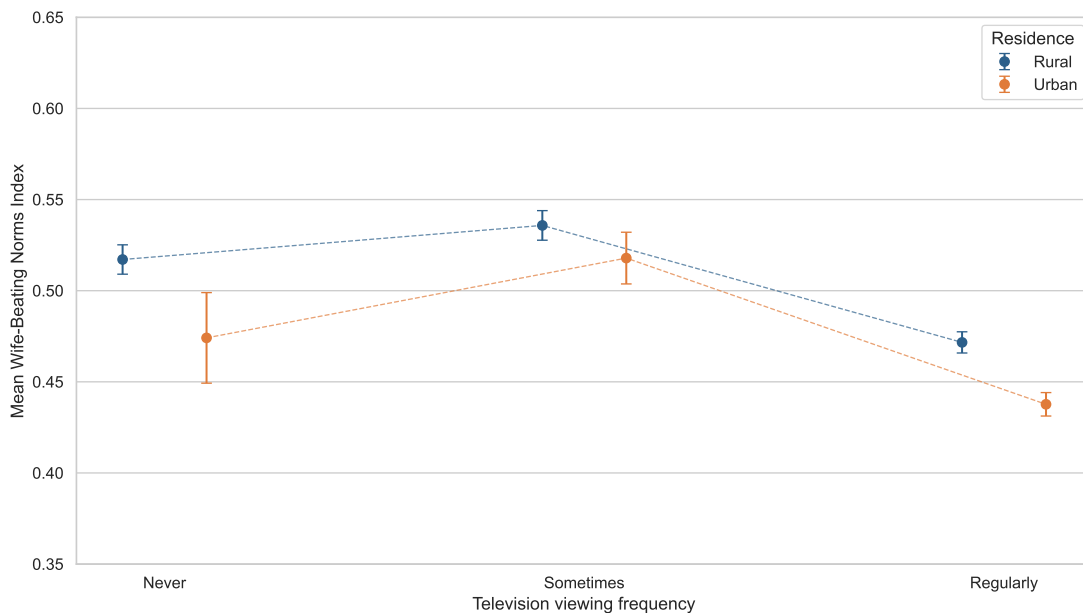


Figure 3: Mean Norms Index by TV Frequency and Residence

Figure 3 disaggregates this pattern by urban and rural residence. Two features stand out. First, the urban–rural gap in the means of Norms Index is small and roughly stable across television-

viewing categories, at around 0.04–0.05 index points. Second, the non-monotonicity is present in both urban and rural series—the index rises slightly from never to sometimes-viewers before falling among regular viewers. The pattern is therefore not specific to either urban or rural areas, suggesting that the underlying compositional differences operate across both contexts.

6 OLS Estimates

Table 4 presents OLS estimates across six specifications, progressively adding controls. The omitted category throughout is women who never watch television.

The coefficient on *Regularly* is negative and statistically significant across all six specifications, ranging from -0.057 in the bivariate specification to -0.043 in the fully controlled Model 6. Although the magnitude attenuates once demographic and socioeconomic controls are introduced, the coefficient remains negative and statistically significant throughout. Relative to the sample mean of 0.484, the Model 6 estimate implies that women who watch television regularly report a Norms Index that is approximately 9% lower than the sample average. The persistence of the coefficient across specifications suggests that the negative association between regular television viewing and Norms Index is robust to the inclusion of a wide range of observed demographic, socioeconomic, and geographic controls. Although this does not rule out bias from unobserved factors, it indicates that the relationship is not driven solely by the observable characteristics included in the model.

The coefficient on *Sometimes* exhibits a more nuanced pattern. It is positive and statistically significant in Models 1 through 5, increasing from 0.018 in the bivariate specification to approximately 0.031 after the inclusion of socioeconomic controls. However, once state fixed effects are added in Model 6, the coefficient reverses sign and becomes negative (-0.012) and statistically significant at the 5% level. This reversal suggests that the positive association observed in earlier specifications is largely driven by cross-state differences. Conditional on residing in the same state, even occasional television viewing is associated with a lower Norms Index.

Among the controls, log consumption per capita is negatively associated with the norms index, although its significance varies across specifications. Age enters with a consistently negative coefficient, indicating that older women tend to report lower levels of Norms Index in their communities. Years of education are strongly and negatively associated with the index across all models, suggesting that more educated women are less likely to perceive wife-beating as a common or normal feature of community life. Urban residence also enters negatively and significantly, indicating lower Norms Index among urban women.

The marital status controls reveal some interesting patterns. Relative to currently married women

living with their spouses, separated or divorced women report substantially higher values of the norms index, suggesting that they perceive wife-beating to be more common within their communities. This may be driven by individual life experiences and possibly correlates with their marital status.⁷

The large increase in R-squared following the inclusion of state fixed effects suggests that cross-state variation accounts for an important share of variation in the Norms Index.

Note: The sample size varies slightly across the six specifications. This is due to missing values. However, the proportion of observations dropped is quite small relative to the overall sample size (about 0.05% once all controls are added), so concerns regarding sample selection do not arise.

⁷For instance, marital separation may be a consequence of violence experienced within the marriage. Left unchecked by the natal family and the in-laws, this may contribute to a higher perceived social acceptance of violence.

Table 4: OLS Estimates: Television Viewing and Norms Index

	(1)	(2)	(3)	(4)	(5)	(6)
Sometimes	0.0180*** (0.00620)	0.0235*** (0.00592)	0.0309*** (0.00632)	0.0325*** (0.00619)	0.0307*** (0.00614)	-0.0124** (0.00603)
Regularly	-0.0568*** (0.00721)	-0.0443*** (0.00656)	-0.0299*** (0.00773)	-0.0259*** (0.00759)	-0.0262*** (0.00714)	-0.0432*** (0.00772)
Log consumption per capita		-0.0208*** (0.00459)	-0.00914* (0.00507)	-0.00733 (0.00471)	-0.00533 (0.00485)	-0.0118*** (0.00320)
Age			-0.000908** (0.000356)	-0.000838** (0.000360)	-0.000612* (0.000357)	-0.000753*** (0.000260)
Years of education (self)			-0.00462*** (0.000646)	-0.00422*** (0.000652)	-0.00316*** (0.000640)	-0.00326*** (0.000369)
Married, spouse absent			-0.0132 (0.00826)	-0.0155* (0.00798)	-0.0175** (0.00773)	-0.00958* (0.00532)
Widowed			0.00286 (0.00826)	0.00360 (0.00823)	0.00468 (0.00812)	-0.00988 (0.00817)
Separated/divorced			0.0555*** (0.0191)	0.0570*** (0.0189)	0.0569*** (0.0189)	0.0320* (0.0176)
Urban (=1)				-0.0191** (0.00827)	-0.0238*** (0.00782)	-0.0292*** (0.00734)
Constant	0.513*** (0.00536)	0.710*** (0.0456)	0.642*** (0.0508)	0.623*** (0.0470)	0.528*** (0.0601)	0.720*** (0.0589)
Observations	39,194	39,187	39,184	39,184	39,171	39,171
R-squared	0.010	0.011	0.015	0.016	0.022	0.210
Caste & Religion Fixed Effects	No	No	No	No	Yes	Yes
State Fixed Effects	No	No	No	No	No	Yes

Note: Dependent variable is the Norms Index (0–1). Omitted television-viewing category is *Never*. Omitted marital-status category is *Married*. Standard errors clustered at the PSU level in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

7 Heterogeneity and Robustness

7.1 Heterogeneity Analysis

Table 5: Heterogeneity Analysis

	Urban-Rural		Education	
	Rural	Urban	Low Education	High Education
Sometimes	-0.0095* (0.0054)	0.0093 (0.0124)	-0.0138 (0.0085)	-0.0142 (0.0126)
Regularly	-0.0253** (0.0097)	-0.0397*** (0.0096)	-0.0376*** (0.0123)	-0.0547*** (0.0135)
Observations	25,813	13,358	21,433	17,741
R^2	0.204	0.246	0.202	0.219

Note: Each column reports estimates from the fully controlled specification (Model 6) estimated separately for the indicated subsample. The omitted television-viewing category is *Never*. All regressions include age, log consumption per capita, marital-status controls, caste and religion fixed effects, and state fixed effects. Low education refers to the subsample where education is less than or equal to median education of the whole sample, and high education where education is above the median. Standard errors clustered at the PSU level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 5 examines whether the relationship between television viewing and Norms Index differs across subgroups. The table reports estimates from the fully controlled specification (Model 6) estimated separately by place of residence and education level.

The negative association between regular television viewing and the Norms Index is present across all four subsamples. However, the magnitude of the association varies. Among rural women, regular television viewing is associated with a 0.025-point lower Norms Index, compared with a 0.040-point lower index among urban women. Similarly, the coefficient is larger among women with above-median education (-0.055) than among women with below-median education (-0.038).

These results suggest that the association between television viewing and perceived community norms regarding wife-beating is not driven by any single subgroup. The estimated coefficients are larger in magnitude among urban and more educated women, although these subgroup regressions do not establish whether the differences across groups are themselves statistically significant.

The coefficient on occasional television viewing is small and statistically insignificant in three of the four subsamples. This pattern is consistent with the main results, where the strongest and most robust association is observed among women who report watching television regularly.

7.2 Robustness Checks

Table 6 examines the sensitivity of the results to alternative specifications. The first robustness check adds spouse's years of schooling as an additional control. This variable was omitted from the baseline specification because it is only observed for women with a living spouse, which reduces the sample size. The second robustness check re-estimates the baseline model using only married women, including those whose spouse is temporarily absent.

The results are highly stable across specifications. The coefficient on regular television viewing remains negative and statistically significant at the 1% level in all three columns, varying only from -0.043 to -0.045. Similarly, the coefficient on occasional television viewing remains negative and statistically significant throughout.

Comparing Columns (2) and (3) is particularly informative. Adding spouse's education necessarily restricts the sample to women with a living spouse. Column (3) isolates the effect of this sample restriction by re-estimating the baseline specification on a similar population of married women without controlling for spouse's education. The near-identical coefficients in Columns (2) and (3) suggest that neither the inclusion of spouse's education nor the associated change in sample composition materially affects the estimated relationship between television viewing and the Norms Index.

Overall, the stability of the estimates across these alternative specifications provides reassurance that the main findings are not sensitive to these modelling choices.

Table 6: Robustness Checks			
	Main Specification	+ Spouse Education	Married Women Only
Sometimes	-0.0124** (0.0060)	-0.0138** (0.0059)	-0.0133** (0.0058)
Regularly	-0.0432*** (0.0077)	-0.0442*** (0.0078)	-0.0448*** (0.0077)
Observations	39,171	36,570	36,643
R^2	0.210	0.213	0.213

Note: The first column reproduces the fully controlled specification from Table 4. Column (2) additionally controls for spouse's years of schooling. Column (3) restricts the sample to married women, including those whose spouse is temporarily absent. The omitted television-viewing category is *Never*. All regressions include age, years of education, log consumption per capita, marital-status controls (where applicable), urban residence, caste and religion fixed effects, and state fixed effects. Standard errors clustered at the PSU level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

8 Causality

Although the regressions control for a range of observable characteristics, women who watch television regularly may still differ from women who never watch television in ways that are not captured by the data.

One concern is that television viewing may be correlated with household attitudes towards women. Households with more progressive views may be more likely to own a television, encourage media consumption, or allow women greater freedom in how they spend their time. Women living in such households may also be less likely to view wife-beating as a normal feature of community life. In that case, part of the negative relationship between television viewing and the Norms Index may reflect underlying household attitudes rather than the effect of television exposure itself.

A second concern is that television viewing may be capturing broader differences in local development. Even within the same state, some villages and neighbourhoods have better infrastructure, greater access to information, and stronger links to urban areas. These factors can increase television exposure, but they may also contribute to less traditional gender norms. As a result, the estimated television coefficients may partly reflect differences in the kinds of communities women live in, rather than the influence of television alone.

A more convincing causal analysis would require variation in television exposure that is unrelated to household attitudes or local development. For example, one could exploit the staggered expansion of television access across regions, differences in signal availability driven by geography, or other sources of variation that affect television viewing without directly affecting perceptions of wife-beating. Such variation would make it possible to compare women who differ in television exposure for reasons largely independent of underlying household culture or community characteristics, thereby strengthening a causal interpretation of the estimated effects.

9 Bound Analysis

To assess the potential importance of omitted variables, I compute Oster bounds for the coefficient on regular television viewing. The approach asks how strongly unobserved factors would need to relate to both television viewing and the Norms Index — relative to the observed controls — in order to eliminate the estimated association entirely. This proportionality parameter is denoted by δ .

Under Oster’s recommended benchmark of $R_{\max}^2 = 1.3R_{\text{full}}^2$, the estimated δ is 1.216. Since Oster (2013) establishes $\delta = 1$ as a defensible upper bound, a value exceeding 1 implies that the result is robust under the recommended specification: unobserved factors would need to bear a stronger relationship to television viewing than all observed controls combined in order to explain away the

estimated coefficient. Under the more conservative assumption $R_{\max}^2 = 1$, δ falls to 0.099, suggesting fragility under the strictest possible bound.